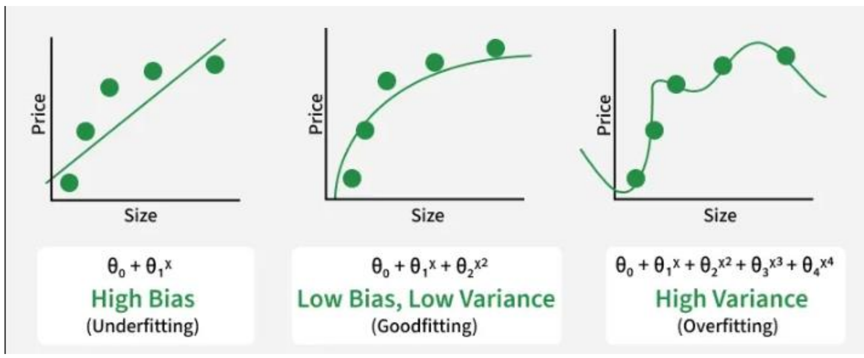
Overfitting vs Underfitting:

- Underfitting is when the model is way too simple to cover all the patterns in real datasets. So, it performs poorly on both the training and new unseen datasets.
- Overfitting is when the model is overly trained to the point that it memorizes the patterns, noise values and values out of the range, which results in a perfect performance on training data but poor performance on new unseen data.
- A good model is one that's not too simple yet not too complex. It should cover the important patterns and not memorize the noises and patterns.



- Causes of overfitting:
  - Insufficient training data: the model doesn't find enough samples to learn properly from
  - Noisy data: datasets with many outliers, errors and irrelative values
  - Complexity of the model: The more complex the model more it will tend to overfit the data.
  - Poor data quality: datasets with many duplicates and missing values
  - Unrepresentative data: the training data doesn't reflect the real world, so its data is biased

- Causes of underfitting:
  - The model is too simple: For example, using a simple linear regression model to predict the trajectory of a rocket (which is non-linear) will result in high error.
  - Insufficient Feature Engineering: If the input features do not contain enough information to predict the target, the model fails to learn.
  - Excessive Regularization: Techniques used to prevent overfitting, such as L1 or L2 regularization, work by constraining the model. They prevent overfitting by penalizing large coefficients. However, if the penalty is too high, you might over-constrain the system. This lead to underfitting, as the model is forced to be too simple to fit the training data.